

APPALA DEEPAK

Software Engineer | Backend & Data Engineering

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building backend services, data-processing pipelines, and AI-powered tools for enterprise workflows. Experienced in Python, Node.js, FastAPI, REST APIs, and LLM integration, with a focus on document processing, data quality, analytics, and workflow automation.

TECHNICAL SKILLS

- **AI / Machine Learning:** LLM Integration, Prompt Engineering, Semantic Search, Embeddings
- **Backend Engineering:** REST API Design, API Development, Data Processing, Backend Development
- **Programming Languages:** Python, JavaScript (ES6), TypeScript, SQL
- **Frameworks & Libraries:** FastAPI, React.js, Node.js, D3.js, Mammoth.js
- **Tools & Infrastructure:** Git, Linux, SQLite, Server-Sent Events, VS Code, Jira, Postman

PROJECTS

Supplementary File Analyzer

Python, FastAPI, LLM Integration, Embeddings

- Built a dashboard tool that ingests zipped article submissions containing the main manuscript and supplementary files (spreadsheets, Word documents, images, and other formats), giving editors a single view to review content and generate LLM-based summaries and integrity checks.
- Developed a spreadsheet quality sub-tool that uses an LLM to generate and execute Python data-quality checks on supporting spreadsheets, producing structured reports on accuracy, consistency, and completeness.
- Created "HypoForge," a sub-tool that generates data-driven hypotheses from supplementary datasets, then validates each hypothesis by writing and running Python tests, and outputs a report of results.
- Added a data-storytelling module that auto-generates charts and narrative summaries from supplementary data, and a semantic search feature that embeds and clusters supporting files so users can retrieve the most relevant file and get an AI-generated answer to natural-language queries.

Nexus Research Intelligence

FastAPI, OpenAlex API, LLM Integration, SQLite

- Built a research-intelligence platform using OpenAlex, Crossref, and publisher catalog data to identify under-published topics and recommend relevant authors.
- Designed a six-stage pipeline for research-field resolution, large-scale work retrieval, LLM-based topic clustering, and market scoring using research momentum, citation velocity, and publisher activity.
- Implemented author ranking, automated acquisition-email generation, and FastAPI-based SSE streaming with SQLite caching for responsive repeat analyses.

EXPERIENCE

Software Engineer

Feb 2024 – Present

Straive, Chennai, Tamil Nadu

- Developed and maintained Node.js services and RESTful APIs for document ingestion, validation, and workflow automation, improving processing efficiency by 35%.
- Implemented a DOCX/DOC/PDF-to-HTML conversion engine using Mammoth.js, extending support to images and structured content across multiple document formats with consistent HTML output.
- Built interactive data visualization dashboards using D3.js to represent operational metrics, workflow status, and submission analytics.
- Developed responsive web interfaces using React, JavaScript, HTML, and CSS to support complex editorial workflows.
- Led rapid POC development, managing a team of 2 engineers and delivering 6+ client-ready POCs within 14-day Agile cycles.
- Collaborated with Product Managers, designers, and backend teams to deliver enterprise-grade solutions, reducing manual handling effort by 40%.

EDUCATION

B.Tech in Computer Science

2020 – 2024

Puducherry Technological University, Puducherry

CGPA: 8.5

ACHIEVEMENTS

- Awarded the Team Synergy Award at Straive for collaboration and improving team productivity.